

Calculus Quick Review

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Preface

These notes are intended as a quick review of the calculus topics with which I am less familiar. They are based on my review of Stewart's *Calculus Early Transcendentals, Metric Edition*, 9th edition.

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1. ■ Calculus I

§1.1. Intermediate Value Theorem

The Intermediate Value Theorem expresses a fundamental property of continuous functions: a continuous graph cannot move from one height to another without passing through every height in between.

Theorem 1.1.1 (Intermediate Value Theorem)

Let f be continuous on the closed interval $[a, b]$, and let N be any number between $f(a)$ and $f(b)$. Then there exists at least one number $c \in [a, b]$ such that

$$f(c) = N.$$

In particular, if N lies strictly between $f(a)$ and $f(b)$, then c may be chosen in (a, b) .

Proof. If $N = f(a)$ or $N = f(b)$, choose $c = a$ or $c = b$, respectively. It therefore remains to consider the case in which N lies strictly between the endpoint values. Without loss of generality, suppose that

$$f(a) < N < f(b).$$

(If $f(b) < N < f(a)$, apply the following argument to $-f$.)

Define

$$S = \{x \in [a, b] : f(x) \leq N\}.$$

The set S is nonempty because $a \in S$, and it is bounded above by b . By the completeness property of the real numbers, S has a least upper bound; let

$$c = \sup S.$$

Continuity at a , together with $f(a) < N$, shows that points just to the right of a belong to S , so $c > a$. Similarly, continuity at b , together with $f(b) > N$, shows that no point sufficiently close to b from the left belongs to S , so $c < b$. Thus $c \in (a, b)$.

We now prove that $f(c) = N$. Suppose first that $f(c) < N$. By continuity at c , there is some $\delta > 0$ such that $f(x) < N$ whenever $|x - c| < \delta$. Since $c < b$, we may choose $x \in (c, b]$ with $|x - c| < \delta$. Then $x \in S$, contradicting the fact that c is an upper bound of S .

Suppose instead that $f(c) > N$. Continuity at c gives some $\delta > 0$ such that $f(x) > N$ whenever $|x - c| < \delta$. Because $c = \sup S$, there must be an $x \in S$ satisfying $c - \delta < x \leq c$; otherwise, $c - \delta$ would be an upper bound of S smaller than c . But this x satisfies both $f(x) \leq N$, since $x \in S$, and $f(x) > N$, by continuity. This is a contradiction.

Neither $f(c) < N$ nor $f(c) > N$ is possible. Therefore $f(c) = N$, as required. ■

§1.2. Derivatives of Trigonometric Functions

Two of the most essential formulas are

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0.$$

By this, and the definition of derivatives, we can obtain

$\frac{d}{dx} \sin x = \cos x$	$\frac{d}{dx} \csc x = -\csc x \cot x$
$\frac{d}{dx} \cos x = -\sin x$	$\frac{d}{dx} \sec x = \sec x \tan x$
$\frac{d}{dx} \tan x = \sec^2 x$	$\frac{d}{dx} \cot x = -\csc^2 x$

Table 1.1.: Derivatives of Trigonometric Functions

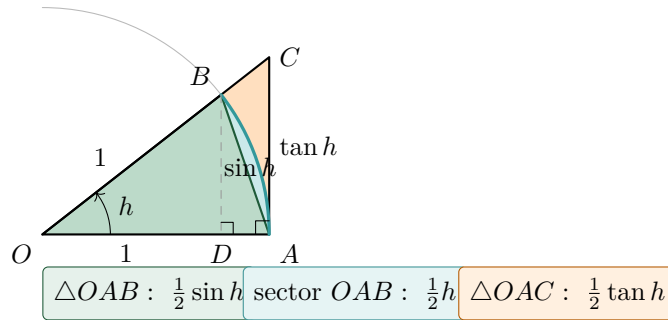
Theorem 1.2.1 (Two fundamental trigonometric limits)

When angles are measured in radians,

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0.$$

Proof. We first prove the geometric inequality needed for the first limit.

Geometric inequality. Let $0 < h < \frac{\pi}{2}$. In the following unit-circle diagram, $OA = OB = 1$, the line AC is tangent to the circle at A , and $\angle AOB = h$.



Indeed, $OA = 1$ and the altitude $BD = \sin h$, so

$$[OAB] = \frac{1}{2}(OA)(BD) = \frac{1}{2} \sin h.$$

The sector has area $h/2$ because its radius is 1, while $AC = \tan h$ gives $[OAC] = \frac{1}{2} \tan h$. Since the three shaded regions are nested, their areas satisfy

$$\underbrace{\frac{1}{2} \sin h}_{[OAB]} \leq \frac{1}{2} h \leq \underbrace{\frac{1}{2} \tan h}_{[OAC]}.$$

Since $\sin h > 0$ and $\cos h > 0$, these inequalities imply

$$\cos h \leq \frac{\sin h}{h} \leq 1.$$

■

The function $\sin h/h$ is even, so for $0 < |h| < \frac{\pi}{2}$ the preceding inequality becomes

$$\cos |h| \leq \frac{\sin h}{h} \leq 1.$$

The same geometric comparison gives $|\sin x| \leq |x|$ near 0. Consequently,

$$0 \leq 1 - \cos |h| = 2 \sin^2 \left(\frac{|h|}{2} \right) \leq \frac{h^2}{2},$$

so $\cos |h| \rightarrow 1$ as $h \rightarrow 0$. The squeeze theorem therefore yields

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1.$$

For the second limit, the identity $\cos h - 1 = -2 \sin^2(h/2)$ gives

$$\frac{\cos h - 1}{h} = -\sin\left(\frac{h}{2}\right) \frac{\sin(h/2)}{h/2}.$$

Moreover, $|\sin(h/2)| \leq |h|/2$, and hence $\sin(h/2) \rightarrow 0$. Together with the first limit, this proves

$$\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = -(0)(1) = 0.$$

■

§1.3. Derivatives of Logarithmic and Inverse Trigonometric Functions

§1.3.i. Logarithmic Functions

Theorem 1.3.1

$$\frac{d}{dx} \log_b x = \frac{1}{x \ln b}.$$

Proof. Let

$$y = \log_b x,$$

where $x > 0$, $b > 0$, and $b \neq 1$. By the definition of a logarithm, this is equivalent to

$$b^y = x.$$

Differentiating both sides implicitly with respect to x gives

$$b^y \ln b \frac{dy}{dx} = 1.$$

Since $b^y = x$, it follows that

$$\frac{dy}{dx} = \frac{1}{b^y \ln b} = \frac{1}{x \ln b}.$$

Therefore,

$$\frac{d}{dx} \log_b x = \frac{1}{x \ln b}.$$

Remark 1.3.1.

$$\frac{d}{dx} b^x = b^x \ln b.$$

■

Theorem 1.3.2

$$\frac{d}{dx} \ln |x| = \frac{1}{x}.$$

Definition 1.3.3. $e = \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.

§1.3.ii. Trigonometric Functions

$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}$
$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$	$\frac{d}{dx} \cot^{-1} x = -\frac{1}{1+x^2}$
$\frac{d}{dx} \sec^{-1} x = \frac{1}{ x \sqrt{x^2-1}}$	$\frac{d}{dx} \csc^{-1} x = -\frac{1}{ x \sqrt{x^2-1}}$

Table 1.2.: Derivatives of inverse trigonometric functions

Theorem 1.3.4

For $-1 < x < 1$,

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}.$$

Proof. Let

$$y = \sin^{-1} x.$$

Since the range of $\sin^{-1}$ is $[-\frac{\pi}{2}, \frac{\pi}{2}]$, this is equivalent to

$$\sin y = x, \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}.$$

Differentiating $\sin y = x$ implicitly with respect to x gives

$$\cos y \frac{dy}{dx} = 1,$$

and hence

$$\frac{dy}{dx} = \frac{1}{\cos y}.$$

On the stated range of y , we have $\cos y \geq 0$. Therefore, using $\sin y = x$ and the identity $\sin^2 y + \cos^2 y = 1$,

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}.$$

Substitution yields

$$\frac{d}{dx} \sin^{-1} x = \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}.$$

■

§1.4. Hyperbolic Functions

Hyperbolic functions are combinations of the exponential functions e^x and e^{-x} . They are analogous to trigonometric functions, but they are associated with the hyperbola $u^2 - v^2 = 1$ rather than the circle $u^2 + v^2 = 1$.

Definition 1.4.1. For every real number x , define

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x}.$$

The reciprocal hyperbolic functions are

$$\operatorname{sech} x = \frac{1}{\cosh x}, \quad \operatorname{csch} x = \frac{1}{\sinh x}, \quad \operatorname{coth} x = \frac{\cosh x}{\sinh x}.$$

Thus $\operatorname{csch} x$ and $\operatorname{coth} x$ are undefined at $x = 0$, whereas the other four functions are defined for every real x .

The following graphs show the characteristic shapes of all six functions. The dashed lines are horizontal or vertical asymptotes.

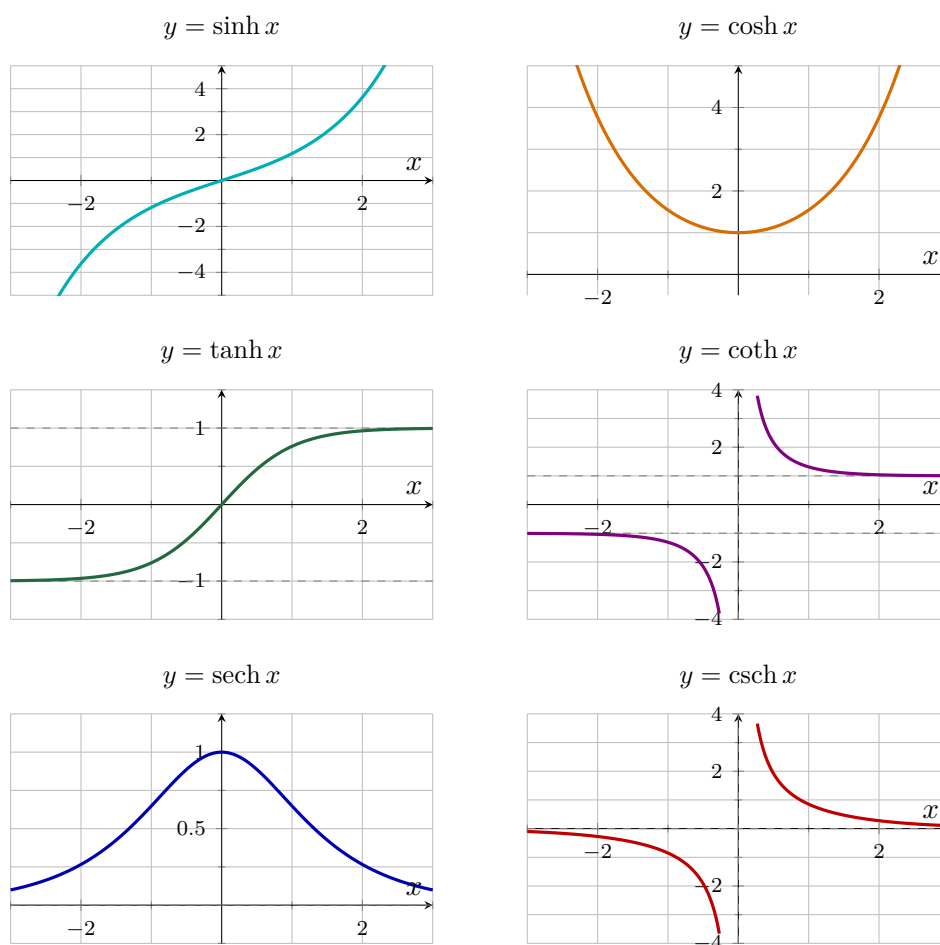


Figure 1.1.: Graphs of the six hyperbolic functions

The functions $\cosh x$ and $\operatorname{sech} x$ are even; the other four are odd. Their ranges are summarized below.

Function	Domain	Range
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$
$\cosh x$	$\mathbb{R}$	$[1, \infty)$
$\tanh x$	$\mathbb{R}$	$(-1, 1)$
$\operatorname{coth} x$	$\mathbb{R} \setminus \{0\}$	$(-\infty, -1) \cup (1, \infty)$
$\operatorname{sech} x$	$\mathbb{R}$	$(0, 1]$
$\operatorname{csch} x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$

Table 1.3.: Domains and ranges of hyperbolic functions

§1.4.i. Hyperbolic Identities

The exponential definitions give the fundamental identity

$$\cosh^2 x - \sinh^2 x = 1.$$

Dividing it by $\cosh^2 x$ or $\sinh^2 x$, respectively, gives

$$1 - \tanh^2 x = \operatorname{sech}^2 x, \quad \coth^2 x - 1 = \operatorname{csch}^2 x.$$

The addition and subtraction formulas are

$$\begin{aligned}\sinh(x \pm y) &= \sinh x \cosh y \pm \cosh x \sinh y, \\ \cosh(x \pm y) &= \cosh x \cosh y \pm \sinh x \sinh y, \\ \tanh(x \pm y) &= \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y},\end{aligned}$$

whenever the denominator in the last expression is nonzero. In particular,

$$\sinh(2x) = 2 \sinh x \cosh x, \quad \cosh(2x) = \cosh^2 x + \sinh^2 x.$$

§1.4.ii. Derivatives of Hyperbolic Functions

For example, differentiating the exponential definition gives

$$\frac{d}{dx} \sinh x = \frac{e^x + e^{-x}}{2} = \cosh x.$$

The remaining formulas follow in the same way, together with the quotient and reciprocal rules.

$(\sinh x)' = \cosh x$	$(\cosh x)' = \sinh x$
$(\tanh x)' = \operatorname{sech}^2 x$	$(\coth x)' = -\operatorname{csch}^2 x$
$(\operatorname{sech} x)' = -\operatorname{sech} x \tanh x$	$(\operatorname{csch} x)' = -\operatorname{csch} x \coth x$

Table 1.4.: Derivatives of hyperbolic functions

§1.4.iii. Inverse Hyperbolic Functions

An inverse hyperbolic function reverses a one-to-one branch of a hyperbolic function. We use the notation $\operatorname{arsinh} x$, $\operatorname{arcosh} x$, and so on; this avoids confusing an inverse function with a reciprocal. For example,

$$y = \operatorname{arsinh} x \iff x = \sinh y.$$

The function $\cosh x$ is restricted to $[0, \infty)$ before it is inverted, while $\coth x$ is inverted separately on its negative and positive branches. The real principal branches are listed below.

Function	Domain	Range
$\operatorname{arsinh} x$	$\mathbb{R}$	$\mathbb{R}$
$\operatorname{arcosh} x$	$[1, \infty)$	$[0, \infty)$
$\operatorname{artanh} x$	$(-1, 1)$	$\mathbb{R}$
$\operatorname{arcoth} x$	$(-\infty, -1) \cup (1, \infty)$	$\mathbb{R} \setminus \{0\}$
$\operatorname{arsech} x$	$(0, 1]$	$[0, \infty)$
$\operatorname{arcsch} x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$

Table 1.5.: Domains and ranges of inverse hyperbolic functions

To solve for an inverse hyperbolic function, write the original function in terms of exponentials and solve the resulting quadratic equation. For example, if $y = \operatorname{arsinh} x$, then

$$x = \frac{e^y - e^{-y}}{2}.$$

Multiplying by $2e^y$ and setting $u = e^y > 0$ gives

$$u^2 - 2xu - 1 = 0.$$

Only the positive root is possible, so

$$e^y = x + \sqrt{x^2 + 1}, \quad \operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1}).$$

The same method gives the logarithmic forms

$$\begin{aligned} \operatorname{arcosh} x &= \ln(x + \sqrt{x^2 - 1}), & x &\geq 1, \\ \operatorname{artanh} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), & |x| &< 1, \\ \operatorname{arcoth} x &= \frac{1}{2} \ln\left(\frac{x+1}{x-1}\right), & |x| &> 1, \\ \operatorname{arsech} x &= \ln\left(\frac{1 + \sqrt{1-x^2}}{x}\right), & 0 < x &\leq 1, \\ \operatorname{arcsch} x &= \operatorname{arsinh}\left(\frac{1}{x}\right) = \ln\left(\frac{1}{x} + \sqrt{1 + \frac{1}{x^2}}\right), & x &\neq 0. \end{aligned}$$

Inverse-function differentiation is often simpler than differentiating the logarithmic formulas. If $y = \operatorname{arsinh} x$, then

$$x = \sinh y, \quad 1 = \cosh y \frac{dy}{dx}.$$

Since $\cosh y = \sqrt{1 + \sinh^2 y} = \sqrt{1 + x^2}$, we obtain

$$\frac{d}{dx} \operatorname{arsinh} x = \frac{1}{\sqrt{1 + x^2}}.$$

Applying the same idea to the other inverse functions gives

$(\operatorname{arsinh} x)' = \frac{1}{\sqrt{1+x^2}}$	$(\operatorname{arcosh} x)' = \frac{1}{\sqrt{x^2-1}}, \quad x > 1$
$(\operatorname{artanh} x)' = \frac{1}{1-x^2}, \quad x < 1$	$(\operatorname{arcoth} x)' = \frac{1}{1-x^2}, \quad x > 1$
$(\operatorname{arsech} x)' = -\frac{1}{x\sqrt{1-x^2}}, \quad 0 < x < 1$	$(\operatorname{arcsch} x)' = -\frac{1}{ x \sqrt{1+x^2}}, \quad x \neq 0$

Table 1.6.: Derivatives of inverse hyperbolic functions

§1.5. Maximum and Minimum Values

Theorem 1.5.1 (The Extreme Value Theorem)

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

Theorem 1.5.2 (Fermat's Theorem)

If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.

Proof. Suppose first that f has a local minimum at c . Then there is some $\delta > 0$ such that

$$f(c+h) \geq f(c) \quad \text{whenever} \quad |h| < \delta.$$

For $0 < h < \delta$, it follows that

$$\frac{f(c+h) - f(c)}{h} \geq 0,$$

whereas for $-\delta < h < 0$, division by the negative number h reverses the inequality and gives

$$\frac{f(c+h) - f(c)}{h} \leq 0.$$

Taking the right-hand limit preserves the first inequality, so

$$f'_+(c) = \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h} \geq 0.$$

Similarly, taking the left-hand limit in the second inequality gives

$$f'_-(c) = \lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h} \leq 0.$$

Because $f'(c)$ exists, the two one-sided derivatives exist and are equal to the derivative:

$$f'_-(c) = f'(c) = f'_+(c).$$

Thus the same number $f'(c)$ is both nonpositive and nonnegative; therefore $f'(c) = 0$.

If f has a local maximum at c , then $f(c+h) \leq f(c)$ for all sufficiently small $|h|$. Hence the inequalities are reversed:

$$f'_+(c) = \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h} \leq 0, \quad f'_-(c) = \lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h} \geq 0.$$

Once again, differentiability at c implies $f'_-(c) = f'(c) = f'_+(c)$, so $f'(c) = 0$. ■

Example 1.5.3 (A local extremum at which the derivative does not exist)

Consider

$$f(x) = |x|.$$

Since $f(x) \geq 0 = f(0)$ for every x , the function has an absolute, and therefore local, minimum at $x = 0$. However,

$$\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1,$$

while

$$\lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1.$$

Thus $f'(0)$ does not exist. This example shows why the assumption that $f'(c)$ exists is necessary in Fermat's Theorem: a function may have a local maximum or minimum at a point where it is not differentiable.

Definition 1.5.4. A **critical number** of a function f is a number c in the domain of f s.t. either $f'(c) = 0$ or $f'(c)$ does not exist.

Thus, we can rephrase Fermat's theorem:

Theorem 1.5.5 (Fermat's theorem (Rephrased))

If f has a local maximum or minimum at c , then c is a critical number of f .

Combining the Extreme Value Theorem with Fermat's Theorem gives the following procedure for finding absolute extrema on a closed interval.

Algorithm 1.1: The Closed Interval Method

Input: A function f that is continuous on $[a, b]$.

Output: The absolute maximum and minimum values of f on $[a, b]$.

- 1 Find every critical number c of f in the open interval (a, b) ; that is, find all c for which $f'(c) = 0$ or $f'(c)$ does not exist;
- 2 Evaluate f at each critical number and at the two endpoints:

$$f(a), f(c_1), f(c_2), \dots, f(b).$$

- 3 Compare these values;
 - 4 The largest value is the absolute maximum value of f on $[a, b]$, and the smallest value is the absolute minimum value;
-

§1.6. The Mean Value Theorem

Theorem 1.6.1 (Rolle's Theorem)

Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$,

Then there is a number c in (a, b) s.t. $f'(c) = 0$.

Proof. Since f is continuous on the closed interval $[a, b]$, the Extreme Value Theorem guarantees that f attains both an absolute maximum and an absolute minimum on $[a, b]$.

If these two extreme values are equal, then f is constant on $[a, b]$. Hence $f'(c) = 0$ for every $c \in (a, b)$, and the result follows.

Now suppose that the absolute maximum and minimum values are not equal. Because $f(a) = f(b)$, at least one of these extreme values differs from the common endpoint value. The point at which that extreme value is attained can therefore be neither a nor b ; thus it is attained at some $c \in (a, b)$. This point c is a local maximum or local minimum of f . Since f is differentiable at every point of (a, b) , Fermat's Theorem gives

$$f'(c) = 0.$$

Therefore, in all cases, there exists some $c \in (a, b)$ such that $f'(c) = 0$. ■

Theorem 1.6.2 (Mean Value Theorem)

Let f be continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) . Then there exists a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define the auxiliary function

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Since f is continuous on $[a, b]$ and differentiable on (a, b) , the function g has the same properties. Moreover,

$$g(a) = f(a)$$

and

$$g(b) = f(b) - \frac{f(b) - f(a)}{b - a}(b - a) = f(a).$$

Thus $g(a) = g(b)$. By Rolle's Theorem, there exists some $c \in (a, b)$ such that $g'(c) = 0$. Since

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a},$$

we obtain

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

Therefore,

$$f'(c) = \frac{f(b) - f(a)}{b - a},$$

as required. ■

Theorem 1.6.3 (Zero Derivative Theorem)

If $f'(x) = 0$ for every $x \in (a, b)$, then f is constant on (a, b) .

Proof. Let $x_1, x_2 \in (a, b)$ with $x_1 < x_2$. Since f is differentiable on (a, b) , it is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2) . By the Mean Value Theorem, there exists some $c \in (x_1, x_2)$ such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c) = 0.$$

Hence $f(x_2) = f(x_1)$. Because x_1 and x_2 were arbitrary, f is constant on (a, b) . ■

Corollary 1.6.4

Let f and g be differentiable on (a, b) . If

$$f'(x) = g'(x)$$

for every $x \in (a, b)$, then f and g differ by a constant; that is, there exists a constant C such that

$$f(x) = g(x) + C$$

for every $x \in (a, b)$.

Proof. Define $h = f - g$. Then h is differentiable on (a, b) , and for every $x \in (a, b)$,

$$h'(x) = f'(x) - g'(x) = 0.$$

By the Zero Derivative Theorem, h is constant on (a, b) . Therefore, there exists a constant C such that

$$f(x) - g(x) = C,$$

or equivalently $f(x) = g(x) + C$, for every $x \in (a, b)$. ■

§1.7. What Derivatives Tell Us about the Shape of a Graph

Derivatives provide information about the shape of a graph. The sign of the first derivative describes where a function rises or falls, while the sign of the second derivative describes how the slope itself changes.

§1.7.i. Increasing and Decreasing Functions

Definition 1.7.1. Let f be defined on an interval I .

1. The function f is **increasing** on I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I .
2. The function f is **decreasing** on I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in I .

Recall that a **critical number** of f is a number c in the domain of f for which $f'(c) = 0$ or $f'(c)$ does not exist. Critical numbers divide the domain into intervals on which the sign of f' can be examined.

Theorem 1.7.2 (Increasing/Decreasing Test)

Suppose that f is continuous on an interval I and differentiable at every interior point of I .

1. If $f'(x) > 0$ throughout the interior of I , then f is increasing on I .
2. If $f'(x) < 0$ throughout the interior of I , then f is decreasing on I .

Proof. Let $x_1 < x_2$ be points in I . By the Mean Value Theorem, there is some $c \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

Since $x_2 - x_1 > 0$, the difference $f(x_2) - f(x_1)$ has the same sign as $f'(c)$. The two conclusions follow. ■

Theorem 1.7.3 (First Derivative Test)

Suppose that c is a critical number of a function f , and that f is continuous at c and differentiable on intervals immediately to the left and right of c .

1. If f' changes from positive to negative at c , then f has a local maximum at c .
2. If f' changes from negative to positive at c , then f has a local minimum at c .
3. If f' has the same sign on both sides of c , then f has no local extremum at c .

Thus the First Derivative Test may be applied by finding the critical numbers, placing them on a number line, determining the sign of f' on each resulting interval, and recording every change of sign.

§1.7.ii. Concavity and the Second Derivative

Definition 1.7.4. Let f be differentiable on an interval I .

1. The graph of f is **concave upward** on I if f' is increasing on I .
2. The graph of f is **concave downward** on I if f' is decreasing on I .

Geometrically, a graph that is concave upward has increasing tangent slopes, while a graph that is concave downward has decreasing tangent slopes. Consequently, if $f''(x) > 0$ on an interval, then f is concave upward there; if $f''(x) < 0$, then f is concave downward there.

Definition 1.7.5. A point $(c, f(c))$ on the graph of a continuous function f is an **inflection point** if the graph changes concavity at c .

The equation $f''(c) = 0$, or the failure of $f''(c)$ to exist, only identifies a possible inflection point. A change of concavity must still be verified on the two sides of c .

Theorem 1.7.6 (Second Derivative Test)

Suppose that $f'(c) = 0$ and that f'' exists on an open interval containing c .

1. If $f''(c) > 0$, then f has a local minimum at c .
2. If $f''(c) < 0$, then f has a local maximum at c .
3. If $f''(c) = 0$, the test is inconclusive: f may have a local maximum, a local minimum, or neither at c .

Proof. Suppose first that $f''(c) > 0$. Since $f'(c) = 0$, the definition of the second derivative gives

$$f''(c) = \lim_{x \rightarrow c} \frac{f'(x) - f'(c)}{x - c} = \lim_{x \rightarrow c} \frac{f'(x)}{x - c} > 0.$$

Hence, sufficiently close to c , the quotient is positive. It follows that $f'(x) < 0$ to the left of c and $f'(x) > 0$ to the right. The First Derivative Test therefore gives a local minimum at c . If $f''(c) < 0$, the signs are reversed, so f has a local maximum at c .

When $f''(c) = 0$, no conclusion follows from this argument. For example, at $c = 0$, the functions x^4 , $-x^4$, and x^3 have, respectively, a local minimum, a local maximum, and no local extremum, although the first and second derivatives at 0 are zero in every case. ■

§1.8. L'Hôpital's Rule

L'Hôpital's Rule is a method for evaluating limits whose direct substitution produces one of the indeterminate forms

$$\frac{0}{0} \quad \text{or} \quad \frac{\infty}{\infty}.$$

The word *indeterminate* is important: these forms do not determine the value of a limit by themselves. The rule replaces the original quotient by the quotient of the derivatives of its numerator and denominator.

Theorem 1.8.1 (L'Hôpital's Rule)

Let f and g be differentiable on an open interval containing a , except possibly at a , and suppose that $g'(x) \neq 0$ near a . Assume either

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$$

or that $|f(x)| \rightarrow \infty$ and $|g(x)| \rightarrow \infty$ as $x \rightarrow a$. If

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L,$$

where L may be finite or infinite, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$$

The same conclusion holds for one-sided limits and for limits as $x \rightarrow \pm\infty$, provided the corresponding hypotheses are satisfied.

For the $0/0$ case, the main idea follows from the Cauchy Mean Value Theorem. If the functions are defined at a by $f(a) = g(a) = 0$, then for each nearby x there is a point c between a and x such that

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}.$$

As $x \rightarrow a$, the point $c \rightarrow a$, which explains why the limit of the derivative quotient determines the original limit.

§1.8.i. How to Apply the Rule

1. Substitute or otherwise verify that the quotient has the form $0/0$ or ∞/∞ .
2. Differentiate the numerator and denominator separately. Do not use the quotient rule.
3. Evaluate the new limit. If it is again indeterminate, the rule may be applied repeatedly, provided its hypotheses continue to hold.

For example,

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x}{1} = 1,$$

because the original quotient has the form $0/0$. Similarly,

$$\lim_{x \rightarrow \infty} \frac{x^2}{e^x} = \lim_{x \rightarrow \infty} \frac{2x}{e^x} = \lim_{x \rightarrow \infty} \frac{2}{e^x} = 0.$$

Other indeterminate forms must first be rewritten. Products of the form $0 \cdot \infty$ can often be expressed as quotients, while differences of the form $\infty - \infty$ may require a common denominator or rationalization. For powers of the forms 0^0 , 1^∞ , or ∞^0 , one usually takes logarithms before applying the rule. For instance, if

$$y = (1 + x)^{1/x},$$

then

$$\ln y = \frac{\ln(1 + x)}{x}, \quad \lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{1/(1 + x)}{1} = 1.$$

Therefore,

$$\lim_{x \rightarrow 0} (1 + x)^{1/x} = e.$$

Remark 1.8.1. L'Hôpital's Rule applies only after an indeterminate form has been verified. It cannot be used directly for a quotient approaching, for example, $0/5$ or $3/0$. It is also unnecessary when algebraic simplification, a standard limit, or the squeeze theorem gives a more direct solution.

2. Calculus II

§2.1. The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus connects the two central operations of calculus: differentiation and integration. Differentiation measures an instantaneous rate of change, while a definite integral accumulates a quantity over an interval. The theorem shows that these operations are, in an important sense, inverses of one another.

§2.1.i. Accumulation Functions

Let f be continuous on $[a, b]$, and define

$$A(x) = \int_a^x f(t) \, dt.$$

The value $A(x)$ is the signed area accumulated under the graph of f from a to x . If x increases by a small amount h , then

$$A(x+h) - A(x) = \int_x^{x+h} f(t) \, dt.$$

For small h , this additional area is approximately $f(x)h$, suggesting that $A'(x) = f(x)$.

Theorem 2.1.1 (Fundamental Theorem of Calculus, Part I)

If f is continuous on $[a, b]$ and

$$A(x) = \int_a^x f(t) \, dt,$$

then A is continuous on $[a, b]$, differentiable on (a, b) , and

$$A'(x) = f(x).$$

Proof. Fix $x \in (a, b)$. For every sufficiently small nonzero h ,

$$\frac{A(x+h) - A(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) \, dt.$$

By the Mean Value Theorem for Integrals, there is a point c_h between x and $x+h$ such that

$$\int_x^{x+h} f(t) \, dt = f(c_h)h.$$

Hence

$$\frac{A(x+h) - A(x)}{h} = f(c_h).$$

As $h \rightarrow 0$, we have $c_h \rightarrow x$. Continuity of f therefore gives

$$A'(x) = \lim_{h \rightarrow 0} f(c_h) = f(x).$$

The continuity of A follows from the same local estimate for the integral over the short interval between x and $x+h$. ■

More generally, the chain rule gives the useful formula

$$\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x))g'(x).$$

For example,

$$\frac{d}{dx} \int_1^{x^2} \sqrt{1+t^3} dt = 2x\sqrt{1+x^6}.$$

§2.1.ii. Evaluating Definite Integrals

An **antiderivative** of f on an interval is a function F such that $F'(x) = f(x)$ throughout that interval. Part II of the theorem turns the evaluation of a definite integral into the problem of finding an antiderivative.

Theorem 2.1.2 (Fundamental Theorem of Calculus, Part II)

If f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Proof. Define

$$A(x) = \int_a^x f(t) dt.$$

By Part I, $A'(x) = f(x)$. Since $F'(x) = f(x)$ as well, the function $F - A$ has derivative zero and is therefore constant. At $x = a$, $A(a) = 0$, so

$$F(x) - A(x) = F(a).$$

Taking $x = b$ and rearranging gives

$$\int_a^b f(t) dt = A(b) = F(b) - F(a).$$

■

The notation

$$F(x)|_a^b \quad \text{or} \quad [F(x)]_a^b$$

means $F(b) - F(a)$. Thus, for example,

$$\int_0^2 (3x^2 - 4x + 1) dx = [x^3 - 2x^2 + x]_0^2 = 2.$$

Remark 2.1.1. The definite integral represents signed accumulation: regions above the horizontal axis contribute positively, while regions below it contribute negatively. Therefore, $\int_a^b f(x) dx$ need not equal the total geometric area between the graph and the axis. To find total area, split the interval at the zeros of f and integrate $|f(x)|$.

§2.2. Strategy for Integration

Integrals containing powers of trigonometric functions are usually handled by saving one factor for the differential and rewriting the remaining factors with a Pythagorean identity. The appropriate choice depends mainly on whether the exponents are odd or even. In the following discussion, m and n are nonnegative integers.

§2.2.i. Integrals of $\sin^m x \cos^n x$

Consider

$$\int \sin^m x \cos^n x \, dx.$$

The identities used most often are

$$\sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x,$$

and the power-reduction formulas

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \quad \cos^2 x = \frac{1 + \cos(2x)}{2}.$$

1. If m is odd, save one factor of $\sin x$, convert the remaining even power of sine into powers of cosine, and use $u = \cos x$:

$$\sin^m x = (1 - \cos^2 x)^{(m-1)/2} \sin x.$$

2. If n is odd, save one factor of $\cos x$, convert the remaining even power of cosine into powers of sine, and use $u = \sin x$:

$$\cos^n x = (1 - \sin^2 x)^{(n-1)/2} \cos x.$$

3. If both m and n are even, repeatedly use the power-reduction formulas. Product-to-sum identities may also simplify the resulting products of multiple-angle functions.

If both exponents are odd, either of the first two substitutions works. For example,

$$\begin{aligned} \int \sin^3 x \cos^2 x \, dx &= \int (1 - \cos^2 x) \cos^2 x \sin x \, dx \\ &= - \int (u^2 - u^4) \, du \quad (u = \cos x) \\ &= -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C. \end{aligned}$$

When both exponents are even, for instance,

$$\begin{aligned} \int \sin^2 x \cos^2 x \, dx &= \frac{1}{4} \int \sin^2(2x) \, dx \\ &= \frac{1}{8} \int (1 - \cos(4x)) \, dx \\ &= \frac{x}{8} - \frac{\sin(4x)}{32} + C. \end{aligned}$$

§2.2.ii. Integrals of $\tan^m x \sec^n x$

Now consider

$$\int \tan^m x \sec^n x \, dx.$$

Here the useful identities and differentials are

$$\tan^2 x = \sec^2 x - 1, \quad \sec^2 x = 1 + \tan^2 x,$$

$$d(\tan x) = \sec^2 x \, dx, \quad d(\sec x) = \sec x \tan x \, dx.$$

1. If n is even and $n \geq 2$, save one factor of $\sec^2 x$, convert the remaining even powers of secant into tangent, and use $u = \tan x$.
2. If m is odd and $n \geq 1$, save one factor of $\sec x \tan x$, convert the remaining even power of tangent into secant, and use $u = \sec x$. This method is especially natural when n is odd.
3. If neither pattern applies, rewrite the integrand using $\tan^2 x = \sec^2 x - 1$. The result may reduce to powers of secant, for which power-reduction formulas or integration by parts can be used. In the special case $n = 0$, rewriting tangent in terms of sine and cosine is often convenient.

For example, because the power of secant is even,

$$\begin{aligned}\int \tan^3 x \sec^4 x \, dx &= \int \tan^3 x (1 + \tan^2 x) \sec^2 x \, dx \\ &= \int u^3 (1 + u^2) \, du \quad (u = \tan x) \\ &= \frac{\tan^4 x}{4} + \frac{\tan^6 x}{6} + C.\end{aligned}$$

On the other hand, when an odd power of tangent is paired with a secant factor,

$$\begin{aligned}\int \tan^3 x \sec^3 x \, dx &= \int (\sec^2 x - 1) \sec^2 x (\sec x \tan x) \, dx \\ &= \int (u^4 - u^2) \, du \quad (u = \sec x) \\ &= \frac{\sec^5 x}{5} - \frac{\sec^3 x}{3} + C.\end{aligned}$$

The central principle in both families is to preserve a factor that matches the differential of the chosen substitution, then use an identity to express every remaining factor in terms of the new variable.

§2.2.iii. Integrals of $\tan x$ and $\sec x$

The antiderivative of tangent follows immediately after writing tangent as a quotient:

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx.$$

Let $u = \cos x$, so that $du = -\sin x \, dx$. Then

$$\begin{aligned}\int \tan x \, dx &= - \int \frac{1}{u} \, du \\ &= -\ln |u| + C \\ &= -\ln |\cos x| + C = \ln |\sec x| + C.\end{aligned}$$

The integral of secant requires a useful algebraic trick. Multiply the integrand by $(\sec x + \tan x)/(\sec x + \tan x)$:

$$\begin{aligned}\int \sec x \, dx &= \int \sec x \frac{\sec x + \tan x}{\sec x + \tan x} \, dx \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx.\end{aligned}$$

With $u = \sec x + \tan x$, we have

$$du = (\sec x \tan x + \sec^2 x) dx.$$

Therefore,

$$\begin{aligned}\int \sec x dx &= \int \frac{1}{u} du \\ &= \ln |u| + C \\ &= \ln |\sec x + \tan x| + C.\end{aligned}$$

These two formulas are worth remembering:

$$\boxed{\int \tan x dx = \ln |\sec x| + C}, \quad \boxed{\int \sec x dx = \ln |\sec x + \tan x| + C}.$$

§2.2.iv. Product-to-Sum Identities

Products of trigonometric functions with different angles can often be integrated more easily after they are converted into sums. The principal product-to-sum identities are

$$\begin{aligned}\sin A \sin B &= \frac{1}{2} [\cos(A - B) - \cos(A + B)], \\ \cos A \cos B &= \frac{1}{2} [\cos(A - B) + \cos(A + B)], \\ \sin A \cos B &= \frac{1}{2} [\sin(A + B) + \sin(A - B)], \\ \cos A \sin B &= \frac{1}{2} [\sin(A + B) - \sin(A - B)].\end{aligned}$$

They follow by adding or subtracting the angle-sum and angle-difference formulas. For example,

$$\cos(A - B) - \cos(A + B) = 2 \sin A \sin B.$$

As an application,

$$\begin{aligned}\int \sin(3x) \cos(2x) dx &= \frac{1}{2} \int [\sin(5x) + \sin x] dx \\ &= -\frac{\cos(5x)}{10} - \frac{\cos x}{2} + C.\end{aligned}$$

Thus, a product-to-sum identity is useful when an integrand contains a product of the form

$$\sin(ax) \sin(bx), \quad \cos(ax) \cos(bx), \quad \text{or} \quad \sin(ax) \cos(bx).$$

The identity separates the product into terms that can be integrated individually.

§2.2.v. Trigonometric Substitution

Trigonometric substitution is useful for integrals involving square roots of quadratic expressions. The substitution is chosen so that a Pythagorean identity removes the radical. For $a > 0$, the three standard patterns are

Expression	Substitution	Identity used
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

For example, if $x = a \sin \theta$, then

$$dx = a \cos \theta \, d\theta \quad \text{and} \quad \sqrt{a^2 - x^2} = a \sqrt{1 - \sin^2 \theta} = a \cos \theta,$$

where $-\pi/2 \leq \theta \leq \pi/2$ is chosen so that $\cos \theta \geq 0$. Similarly, suitable choices of θ give

$$\sqrt{a^2 + x^2} = a \sec \theta \quad \text{when } x = a \tan \theta,$$

and

$$\sqrt{x^2 - a^2} = a \tan \theta \quad \text{when } x = a \sec \theta.$$

Choosing an interval on which the relevant trigonometric functions have the correct signs avoids introducing unnecessary absolute values.

The usual procedure is:

1. Identify which of the three quadratic expressions occurs.
2. Make the corresponding substitution and replace dx .
3. Use a Pythagorean identity to eliminate the square root.
4. Evaluate the resulting trigonometric integral.
5. Rewrite the answer in terms of x .

For instance, consider

$$\int \sqrt{9 - x^2} \, dx.$$

Set $x = 3 \sin \theta$, with $-\pi/2 \leq \theta \leq \pi/2$. Then

$$dx = 3 \cos \theta \, d\theta, \quad \sqrt{9 - x^2} = 3 \cos \theta.$$

Therefore,

$$\begin{aligned} \int \sqrt{9 - x^2} \, dx &= 9 \int \cos^2 \theta \, d\theta \\ &= \frac{9}{2} \int (1 + \cos 2\theta) \, d\theta \\ &= \frac{9}{2} \theta + \frac{9}{4} \sin 2\theta + C. \end{aligned}$$

Since $\sin \theta = x/3$, we have

$$\theta = \sin^{-1}\left(\frac{x}{3}\right), \quad \cos \theta = \frac{\sqrt{9 - x^2}}{3}.$$

Thus

$$\sin 2\theta = 2 \sin \theta \cos \theta = \frac{2x\sqrt{9 - x^2}}{9},$$

and the final result is

$$\boxed{\int \sqrt{9 - x^2} \, dx = \frac{x}{2} \sqrt{9 - x^2} + \frac{9}{2} \sin^{-1}\left(\frac{x}{3}\right) + C.}$$

A right triangle provides a convenient way to return to x . For $x = a \sin \theta$, use opposite side x , hypotenuse a , and adjacent side $\sqrt{a^2 - x^2}$. For $x = a \tan \theta$, use opposite side x , adjacent side a , and hypotenuse $\sqrt{a^2 + x^2}$. For $x = a \sec \theta$ with $x > a$, use hypotenuse x , adjacent side a , and opposite side $\sqrt{x^2 - a^2}$.

§2.2.vi. Integration of Rational Functions by Partial Fractions

A **rational function** is a quotient of polynomials,

$$R(x) = \frac{P(x)}{Q(x)}, \quad Q(x) \neq 0.$$

Partial fraction decomposition rewrites a complicated rational function as a sum of simpler fractions whose antiderivatives are known. The method applies after the denominator has been factored over the real numbers.

Preliminary Steps

First compare the degrees of the two polynomials. If $\deg P \geq \deg Q$, use polynomial long division to write

$$\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}, \quad \deg R < \deg Q.$$

Only the proper rational function $R(x)/Q(x)$ is then decomposed. Next, factor $Q(x)$ completely into linear and irreducible quadratic factors.

The form of the decomposition is determined by those factors:

1. For each distinct linear factor $ax + b$, include

$$\frac{A}{ax + b}.$$

2. For a repeated linear factor $(ax + b)^k$, include one term for every power up to k :

$$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_k}{(ax + b)^k}.$$

3. For each irreducible quadratic factor $ax^2 + bx + c$, include a linear numerator:

$$\frac{Ax + B}{ax^2 + bx + c}.$$

4. For a repeated irreducible quadratic factor $(ax^2 + bx + c)^k$, include

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_kx + B_k}{(ax^2 + bx + c)^k}.$$

The numerators must have degree strictly less than the corresponding denominator factors. Thus constants are used above linear factors, while linear polynomials are used above quadratic factors.

Finding the Coefficients

After writing the proposed decomposition, multiply both sides by the common denominator. The coefficients can then be found by substituting convenient values of x and comparing coefficients of equal powers of x .

For example, consider

$$\int \frac{2x^2 + 3x - 2}{x(x - 1)(x + 2)} dx.$$

The denominator consists of three distinct linear factors, so write

$$\frac{2x^2 + 3x - 2}{x(x-1)(x+2)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+2}.$$

Multiplying by $x(x-1)(x+2)$ gives

$$2x^2 + 3x - 2 = A(x-1)(x+2) + Bx(x+2) + Cx(x-1).$$

Substituting the zeros of the denominator factors yields

$$\begin{array}{lll} x = 0 : & -2 = -2A, & A = 1, \\ x = 1 : & 3 = 3B, & B = 1, \\ x = -2 : & 0 = 6C, & C = 0. \end{array}$$

Therefore,

$$\frac{2x^2 + 3x - 2}{x(x-1)(x+2)} = \frac{1}{x} + \frac{1}{x-1},$$

and hence

$$\begin{aligned} \int \frac{2x^2 + 3x - 2}{x(x-1)(x+2)} dx &= \int \left(\frac{1}{x} + \frac{1}{x-1} \right) dx \\ &= \ln|x| + \ln|x-1| + C. \end{aligned}$$

Quadratic Factors

Terms with irreducible quadratic denominators usually produce both logarithms and inverse tangent functions. The inverse tangent appears because

$$\frac{d}{du} \tan^{-1} u = \frac{1}{1+u^2}.$$

Consequently,

$$\int \frac{1}{1+u^2} du = \tan^{-1} u + C.$$

More generally, for $a > 0$, let $u = (x-h)/a$. Then $x-h = au$ and $dx = a du$, so

$$\begin{aligned} \int \frac{1}{(x-h)^2 + a^2} dx &= \int \frac{a}{a^2 u^2 + a^2} du \\ &= \frac{1}{a} \int \frac{1}{1+u^2} du \\ &= \frac{1}{a} \tan^{-1} u + C. \end{aligned}$$

Thus the standard formula is

$$\boxed{\int \frac{1}{(x-h)^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x-h}{a} \right) + C.}$$

To integrate a fraction with an irreducible quadratic denominator, first separate the numerator into a multiple of the derivative of the denominator and a constant remainder. The derivative part produces a logarithm; after completing the square, the remainder produces an inverse tangent.

For example, consider

$$\int \frac{3x+5}{x^2+4x+8} dx.$$

The derivative of the denominator is $2x + 4$. Rewrite the numerator as

$$3x + 5 = \frac{3}{2}(2x + 4) - 1.$$

The integral therefore separates into

$$\int \frac{3x + 5}{x^2 + 4x + 8} dx = \frac{3}{2} \int \frac{2x + 4}{x^2 + 4x + 8} dx - \int \frac{1}{x^2 + 4x + 8} dx.$$

For the first term, the substitution $v = x^2 + 4x + 8$ gives

$$\frac{3}{2} \int \frac{2x + 4}{x^2 + 4x + 8} dx = \frac{3}{2} \ln(x^2 + 4x + 8).$$

For the second term, complete the square:

$$x^2 + 4x + 8 = (x + 2)^2 + 4 = (x + 2)^2 + 2^2.$$

The standard formula applies with $h = -2$ and $a = 2$, giving

$$\int \frac{1}{(x + 2)^2 + 2^2} dx = \frac{1}{2} \tan^{-1}\left(\frac{x + 2}{2}\right) + C.$$

Equivalently, this can be seen directly by setting $u = (x + 2)/2$. Then $dx = 2 du$, and

$$\begin{aligned} \int \frac{1}{(x + 2)^2 + 4} dx &= \int \frac{2}{4u^2 + 4} du \\ &= \frac{1}{2} \int \frac{1}{1 + u^2} du \\ &= \frac{1}{2} \tan^{-1} u + C. \end{aligned}$$

Combining the logarithmic and inverse tangent parts yields

$$\boxed{\int \frac{3x + 5}{x^2 + 4x + 8} dx = \frac{3}{2} \ln(x^2 + 4x + 8) - \frac{1}{2} \tan^{-1}\left(\frac{x + 2}{2}\right) + C.}$$

In summary, the partial-fractions procedure is:

1. Perform long division if the rational function is improper.
2. Factor the denominator completely over the real numbers.
3. Write all partial-fraction terms, including every repeated power.
4. Clear denominators and solve for the unknown coefficients.
5. Integrate the simpler terms separately.

§2.3. Improper Integrals

A definite integral is called **improper** when either its interval of integration is unbounded or its integrand becomes unbounded at one or more points of the interval. Such an integral is not evaluated by substituting infinity into an antiderivative. Instead, it is defined as a limit of proper integrals.

§2.3.i. Infinite Intervals

Suppose that f is continuous on $[a, \infty)$. We define

$$\int_a^\infty f(x) \, dx = \lim_{t \rightarrow \infty} \int_a^t f(x) \, dx,$$

provided that the limit exists and is finite. In that case the improper integral **converges**; otherwise, it **diverges**. Similarly,

$$\int_{-\infty}^b f(x) \, dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) \, dx.$$

An integral over the entire real line must be split at an arbitrary finite number c :

$$\int_{-\infty}^\infty f(x) \, dx = \int_{-\infty}^c f(x) \, dx + \int_c^\infty f(x) \, dx.$$

It converges only when both integrals on the right converge. The value does not depend on the choice of c . In particular, one must not replace this definition by the single symmetric limit $\lim_{t \rightarrow \infty} \int_{-t}^t f(x) \, dx$, which may exist even when the improper integral diverges.

For example,

$$\begin{aligned} \int_1^\infty \frac{1}{x^2} \, dx &= \lim_{t \rightarrow \infty} \int_1^t x^{-2} \, dx \\ &= \lim_{t \rightarrow \infty} \left[-\frac{1}{x} \right]_1^t = \lim_{t \rightarrow \infty} \left(1 - \frac{1}{t} \right) = 1. \end{aligned}$$

Thus the integral converges. By contrast,

$$\int_1^\infty \frac{1}{x} \, dx = \lim_{t \rightarrow \infty} \ln t = \infty,$$

so this integral diverges.

§2.3.ii. Unbounded Integrands

If f is continuous on $(a, b]$ and becomes unbounded as $x \rightarrow a^+$, define

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow a^+} \int_t^b f(x) \, dx.$$

If instead f becomes unbounded as $x \rightarrow b^-$, define

$$\int_a^b f(x) \, dx = \lim_{t \rightarrow b^-} \int_a^t f(x) \, dx.$$

As before, convergence requires the corresponding limit to exist and be finite.

If f is unbounded at an interior point $c \in (a, b)$, the integral must be split there:

$$\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx.$$

The original integral converges if and only if both one-sided improper integrals converge. Each improper endpoint or interior singularity must be handled with its own limit.

For instance,

$$\begin{aligned} \int_0^1 \frac{1}{\sqrt{x}} \, dx &= \lim_{t \rightarrow 0^+} \int_t^1 x^{-1/2} \, dx \\ &= \lim_{t \rightarrow 0^+} [2\sqrt{x}]_t^1 = \lim_{t \rightarrow 0^+} (2 - 2\sqrt{t}) = 2. \end{aligned}$$

Although the integrand is unbounded near 0, the integral converges.

§2.3.iii. The p -Integral Test

The basic benchmark for many convergence questions is the following result.

Theorem 2.3.1 (The p -integral test)

For a real number p ,

$$\int_1^{\infty} \frac{1}{x^p} dx$$

converges if and only if $p > 1$. Also,

$$\int_0^1 \frac{1}{x^p} dx$$

converges if and only if $p < 1$.

Proof. If $p \neq 1$, then

$$\int x^{-p} dx = \frac{x^{1-p}}{1-p} + C.$$

Therefore,

$$\int_1^t x^{-p} dx = \frac{t^{1-p} - 1}{1-p}.$$

As $t \rightarrow \infty$, this expression has a finite limit precisely when $1 - p < 0$, or $p > 1$. When $p = 1$, the integral becomes $\int_1^t \frac{1}{x} dx = \ln t$, which diverges.

Similarly,

$$\int_t^1 x^{-p} dx = \frac{1 - t^{1-p}}{1-p}.$$

As $t \rightarrow 0^+$, this expression has a finite limit precisely when $1 - p > 0$, or $p < 1$. The case $p = 1$ again gives a divergent logarithm. ■

The corresponding p -series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

has the same convergence condition: it converges if and only if $p > 1$. This connection follows from the integral test for infinite series. The two objects are different—one is an integral and the other is a sum—but they provide matching benchmark examples.

§2.3.iv. Comparison Tests

When an improper integral is difficult to evaluate exactly, its convergence can often be determined by comparing its integrand with a simpler benchmark, usually a power function.

Theorem 2.3.2 (Direct comparison test for improper integrals)

Suppose that f and g are continuous and satisfy

$$0 \leq f(x) \leq g(x)$$

for all sufficiently large x . Then:

1. If $\int_a^\infty g(x) \, dx$ converges, then $\int_a^\infty f(x) \, dx$ converges.
2. If $\int_a^\infty f(x) \, dx$ diverges, then $\int_a^\infty g(x) \, dx$ diverges.

The analogous statements hold near a finite improper endpoint.

The direction of comparison is important: an upper bound by a convergent integral proves convergence, while a lower bound by a divergent integral proves divergence.

For example, consider

$$\int_1^\infty \frac{1}{x^2 + 4} \, dx.$$

For $x \geq 1$,

$$0 \leq \frac{1}{x^2 + 4} \leq \frac{1}{x^2}.$$

Since $\int_1^\infty x^{-2} \, dx$ converges, the direct comparison test shows that the given integral converges.

As a divergence example, for $x \geq 1$,

$$\frac{1}{\sqrt{x^2 + 1}} \geq \frac{1}{\sqrt{2}x} = \frac{1}{\sqrt{2}} \frac{1}{x}.$$

Because $\int_1^\infty \frac{1}{\sqrt{2}x} \, dx$ diverges, so does

$$\int_1^\infty \frac{1}{\sqrt{x^2 + 1}} \, dx.$$

Theorem 2.3.3 (Limit comparison test for improper integrals)

Let f and g be positive continuous functions for all sufficiently large x . If

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L, \quad 0 < L < \infty,$$

then

$$\int_a^\infty f(x) \, dx \quad \text{and} \quad \int_a^\infty g(x) \, dx$$

either both converge or both diverge. An analogous result holds when x approaches a finite improper endpoint.

Proof. Choose positive constants m and M with $m < L < M$. From the definition of the limit, for all sufficiently large x ,

$$m < \frac{f(x)}{g(x)} < M.$$

Hence

$$mg(x) < f(x) < Mg(x).$$

Applying the direct comparison test in both directions proves the result. ■

For example, let

$$f(x) = \frac{3x+1}{x^3+2} \quad \text{and} \quad g(x) = \frac{1}{x^2}.$$

Then

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{x^2(3x+1)}{x^3+2} = 3.$$

Since 3 is positive and finite, and $\int_1^\infty x^{-2} dx$ converges, the limit comparison test shows that

$$\int_1^\infty \frac{3x+1}{x^3+2} dx$$

also converges.

§2.3.v. Absolute Convergence

For integrands that change sign, comparison tests are usually applied to the absolute value.

Theorem 2.3.4 (Absolute convergence)

If

$$\int_a^\infty |f(x)| dx$$

converges, then

$$\int_a^\infty f(x) dx$$

converges.

The converse need not hold. For example,

$$\int_1^\infty \frac{\sin x}{x} dx$$

converges, but it does not converge absolutely. Such an integral is said to be **conditionally convergent**.

In practice, the main steps are:

1. Identify every infinite endpoint and every point where the integrand is unbounded.
2. Split the integral so that each improper feature has a separate one-sided limit.
3. Evaluate those limits directly when possible.
4. If direct evaluation is difficult, compare the integrand with a known benchmark such as $1/x^p$.
5. Declare the original integral convergent only when every required limit exists and is finite.

§2.4. Arc Length

The length of a straight line segment follows from the Pythagorean theorem. To find the length of a curved graph, we approximate the curve by many short line segments and then take a limit. This process turns a geometric length into a definite integral.

§2.4.i. Derivation of the Arc-Length Formula

Let f have a continuous derivative on $[a, b]$, and consider the graph $y = f(x)$. Choose a partition

$$a = x_0 < x_1 < \cdots < x_n = b$$

and join consecutive points

$$P_i = (x_i, f(x_i))$$

by straight line segments. The diagram below shows both the smooth curve and its polygonal approximation. One chord is enlarged conceptually into a right triangle whose horizontal and vertical legs are Δx_i and Δy_i .

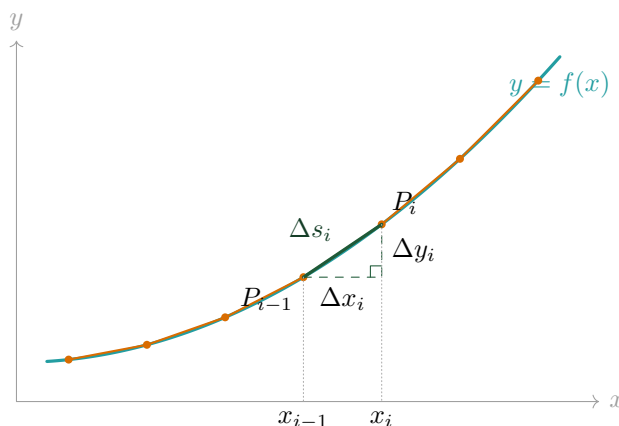


Figure 2.1.: Approximating a smooth curve by chords. Each chord is the hypotenuse of a right triangle with legs Δx_i and Δy_i .

By the Pythagorean theorem, the length of the segment from P_{i-1} to P_i is

$$\begin{aligned} \Delta s_i &= \sqrt{(x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2} \\ &= \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} \\ &= \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2} \Delta x_i, \end{aligned}$$

where $\Delta x_i = x_i - x_{i-1} > 0$ and $\Delta y_i = f(x_i) - f(x_{i-1})$. Thus the length of the polygonal approximation is

$$L_n = \sum_{i=1}^n \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2} \Delta x_i.$$

By the Mean Value Theorem, for each i there is some $c_i \in (x_{i-1}, x_i)$ such that

$$\frac{\Delta y_i}{\Delta x_i} = f'(c_i).$$

Consequently,

$$L_n = \sum_{i=1}^n \sqrt{1 + (f'(c_i))^2} \Delta x_i.$$

As the width of the largest subinterval approaches zero, these polygonal paths follow the curve more and more closely. The sum on the right is a Riemann sum for $\sqrt{1 + (f'(x))^2}$. This motivates, and under the assumption that f' is continuous rigorously gives, the following formula.

Theorem 2.4.1 (Arc length of a graph)

If f' is continuous on $[a, b]$, then the length of the graph $y = f(x)$ from $x = a$ to $x = b$ is

$$L = \int_a^b \sqrt{1 + (f'(x))^2} \, dx.$$

The differential form of the same argument is also useful. Over a very small displacement, the curve is nearly a line segment, so

$$(ds)^2 = (dx)^2 + (dy)^2.$$

Since $dy = f'(x) \, dx$, we obtain

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx.$$

Integrating ds from a to b yields the arc-length formula. The square root appears because each infinitesimal piece combines a horizontal change and a vertical change by the Pythagorean theorem.

§2.4.ii. Other Forms of the Formula

If a curve is more naturally written as $x = g(y)$, where g' is continuous for $c \leq y \leq d$, the roles of x and y may be reversed:

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \, dy.$$

For a parametrized curve

$$x = x(t), \quad y = y(t), \quad \alpha \leq t \leq \beta,$$

with continuous derivatives and no repeated tracing, the small changes are $dx = x'(t) \, dt$ and $dy = y'(t) \, dt$. Therefore,

$$ds = \sqrt{(x'(t))^2 + (y'(t))^2} \, dt,$$

and hence

$$L = \int_{\alpha}^{\beta} \sqrt{(x'(t))^2 + (y'(t))^2} \, dt.$$

This is the integral of the speed of a particle moving along the curve. In particular, if $x(t) = t$ and $y(t) = f(t)$, it reduces to the formula for the graph of a function.

§2.5. Area of a Surface of Revolution

When a plane curve is revolved about a line, it sweeps out a **surface of revolution**. Its surface area depends on both the length of each small piece of the curve and its distance from the axis of rotation. The basic idea is to divide the curve into short segments, revolve each segment to form a thin conical frustum, and then pass to a limit.

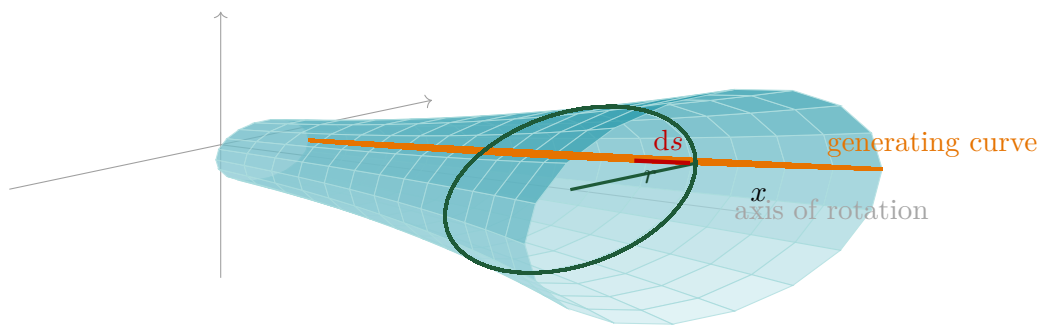


Figure 2.2.: A curve revolved about the x -axis. At each point, a small arc of length ds travels around a circle of radius r , sweeping out a narrow band of area approximately $2\pi r ds$.

§2.5.i. Derivation of the Formula

Let $f(x) \geq 0$ and suppose that f' is continuous on $[a, b]$. Choose a partition

$$P: \quad a = x_0 < x_1 < \cdots < x_n = b$$

and replace the graph of f by the polygonal path through the points

$$P_i = (x_i, y_i), \quad y_i = f(x_i).$$

For each i , join P_{i-1} to P_i by a straight chord and denote its length by

$$c_i = \|P_i - P_{i-1}\|.$$

When this chord is revolved about the x -axis, it generates one conical frustum. Let A_i be the lateral area of that frustum and define

$$A(P) = \sum_{i=1}^n A_i.$$

Thus $A(P)$ is the area of the polygonal surface made from all the frusta. The surface area of the smooth surface is obtained by refining the partition:

$$S = \lim_{\|P\| \rightarrow 0} A(P), \quad \|P\| = \max_{1 \leq i \leq n} \Delta x_i, \quad \Delta x_i = x_i - x_{i-1}.$$

The calculation below proves that this limit exists and identifies it as an integral. Notice that the approximation is built from straight chords, not from the original curved pieces. Each piece in Figure 2.3 is a truncated cone between two nearby circular cross-sections.

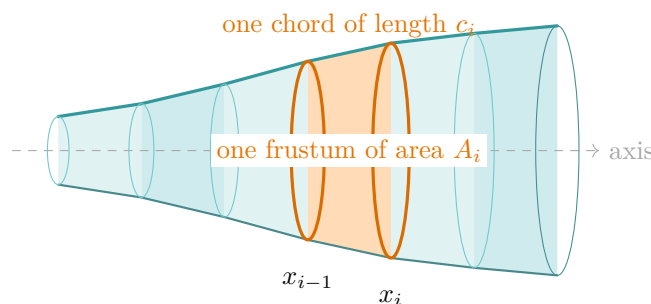


Figure 2.3.: The polygonal curve produces many adjacent conical frusta. Their circular edges match, so the sum $A(P) = A_1 + \cdots + A_n$ approximates the entire surface area.

The notation used in the proof has the following fixed meaning:

$$\begin{aligned}
 P_i &= \text{a point on the original graph,} \\
 c_i &= \text{the straight chord length from } P_{i-1} \text{ to } P_i, \\
 A_i &= \text{the lateral area of the frustum generated by that chord,} \\
 A(P) &= \sum_{i=1}^n A_i, \text{ the polygonal surface area,} \\
 S &= \lim_{\|P\| \rightarrow 0} A(P), \text{ the smooth surface area.}
 \end{aligned}$$

Step 1: the exact area of one frustum. For the i -th chord, let

$$r_{i-1} = f(x_{i-1}), \quad r_i = f(x_i),$$

The two radii are r_{i-1} and r_i , while the slant height is the chord length c_i . Therefore the exact lateral area is

$$A_i = \pi(r_{i-1} + r_i)c_i.$$

This formula can be derived from the lateral area $\pi r \ell$ of a cone. For example, suppose $r_i > r_{i-1}$. Extend the sloping sides of the frustum until they meet at the apex of a cone. If L_i and L_{i-1} are the slant heights from the apex to the two circular edges, similarity of triangles gives

$$\frac{L_i}{r_i} = \frac{L_{i-1}}{r_{i-1}}, \quad L_i - L_{i-1} = c_i.$$

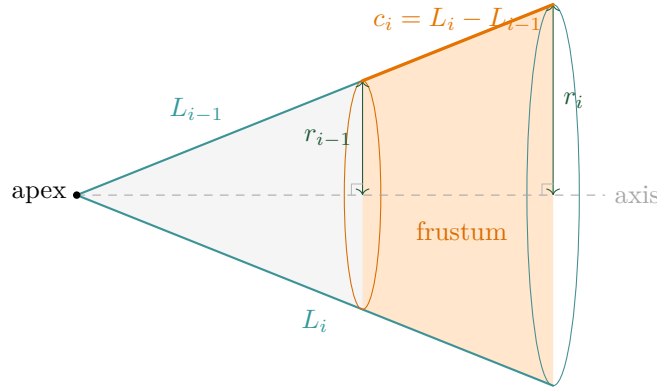


Figure 2.4.: An axial cross-section of the frustum extended to a common apex. The small and large right triangles are similar, so their slant heights and radii are proportional. The frustum's slant height is the difference $c_i = L_i - L_{i-1}$.

Solving these equations yields

$$L_i = \frac{r_i c_i}{r_i - r_{i-1}}, \quad L_{i-1} = \frac{r_{i-1} c_i}{r_i - r_{i-1}}.$$

The frustum is the large cone with the small cone removed, so

$$\begin{aligned}
 A_i &= \pi r_i L_i - \pi r_{i-1} L_{i-1} \\
 &= \pi c_i \frac{r_i^2 - r_{i-1}^2}{r_i - r_{i-1}} \\
 &= \pi(r_i + r_{i-1})c_i.
 \end{aligned}$$

The same result holds if $r_i < r_{i-1}$. If the radii are equal, the frustum is a cylinder and the formula reduces to $2\pi r_i c_i$, as expected. Equivalently,

$$\bar{r}_i = \frac{r_{i-1} + r_i}{2}, \quad A_i = 2\pi \bar{r}_i c_i.$$

Thus the area of one frustum is exactly

$$\underbrace{2\pi \bar{r}_i}_{\text{circumference at average radius}} \times \underbrace{c_i}_{\text{chord length}}.$$

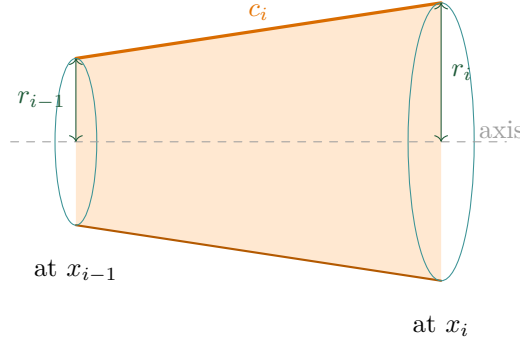


Figure 2.5.: A short chord produces a conical frustum. Its lateral area is $\pi(r_{i-1} + r_i)c_i$, or circumference at the average radius times slant height.

Step 2: express the chord length in terms of f' . Set

$$\Delta x_i = x_i - x_{i-1}, \quad \Delta y_i = f(x_i) - f(x_{i-1}).$$

The slant height of the frustum is exactly the length of the chord, so the Pythagorean theorem gives

$$\begin{aligned} c_i &= \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} \\ &= \sqrt{1 + \left(\frac{\Delta y_i}{\Delta x_i}\right)^2} \Delta x_i. \end{aligned}$$

Because f is continuous on $[x_{i-1}, x_i]$ and differentiable in its interior, the Mean Value Theorem supplies a point $\xi_i \in (x_{i-1}, x_i)$ such that

$$\frac{\Delta y_i}{\Delta x_i} = f'(\xi_i).$$

Therefore,

$$c_i = \sqrt{1 + (f'(\xi_i))^2} \Delta x_i.$$

This is an exact formula for the chord length. It does not assert that c_i is the length of the curved portion of the graph. At this stage we are computing the area of the polygonal approximation, so only the straight chord is needed. Substituting this into the exact frustum formula gives

$$A_i = 2\pi \left(\frac{f(x_{i-1}) + f(x_i)}{2} \right) \sqrt{1 + (f'(\xi_i))^2} \Delta x_i.$$

Hence the area of the entire polygonal surface is

$$A(P) = 2\pi \sum_{i=1}^n \left(\frac{f(x_{i-1}) + f(x_i)}{2} \right) \sqrt{1 + (f'(\xi_i))^2} \Delta x_i.$$

Step 3: pass from the sum to an integral. The only remaining issue is the radius. The exact frustum formula uses the average endpoint radius

$$\bar{r}_i = \frac{f(x_{i-1}) + f(x_i)}{2},$$

whereas a standard Riemann sum evaluates every factor at one sample point. On a very short subinterval, however, $f(x_{i-1})$, $f(\xi_i)$, and $f(x_i)$ are all close. Therefore $\bar{r}_i$ may be replaced by $f(\xi_i)$ in the limit. The following estimate makes this precise.

Define

$$q(x) = \sqrt{1 + (f'(x))^2}.$$

Since f' is continuous on the closed interval $[a, b]$, both q and

$$f(x)q(x) = f(x)\sqrt{1 + (f'(x))^2}$$

are continuous and therefore Riemann integrable. If the average radius in the preceding sum were replaced by $f(\xi_i)$, we would obtain the ordinary Riemann sum

$$2\pi \sum_{i=1}^n f(\xi_i)q(\xi_i) \Delta x_i,$$

which converges to

$$2\pi \int_a^b f(x)q(x) dx.$$

It remains to justify replacing the average endpoint radius by $f(\xi_i)$. Because f is continuous on the compact interval $[a, b]$, it is uniformly continuous. Let

$$\omega_f(\delta) = \sup\{|f(u) - f(v)| : u, v \in [a, b], |u - v| \leq \delta\}$$

be its modulus of continuity. Since x_{i-1} , ξ_i , and x_i all lie in an interval of width at most $\|P\|$,

$$\left| \frac{f(x_{i-1}) + f(x_i)}{2} - f(\xi_i) \right| \leq \omega_f(\|P\|).$$

Also, q is continuous on $[a, b]$, so it has a finite maximum M . The difference between the frustum sum and the ordinary Riemann sum is thus at most

$$\begin{aligned} 2\pi \sum_{i=1}^n \omega_f(\|P\|)M \Delta x_i \\ = 2\pi M(b-a)\omega_f(\|P\|), \end{aligned}$$

which approaches 0 as $\|P\| \rightarrow 0$. Consequently,

$$\begin{aligned} S &= \lim_{\|P\| \rightarrow 0} A(P) \\ &= 2\pi \int_a^b f(x)\sqrt{1 + (f'(x))^2} dx. \end{aligned}$$

Only after taking this limit do we introduce the arc-length differential ds . It abbreviates the limiting factor

$$ds = \sqrt{1 + (f'(x))^2} dx.$$

Thus the final formula is often remembered symbolically as

$$dS = 2\pi f(x) ds.$$

This notation describes the limiting integral. It does not say that the finite chord length c_i is equal to the length of the corresponding curved piece. Geometrically, it says that an infinitesimally narrow surface band has area equal to its circumference $2\pi f(x)$ times its infinitesimal slant width ds .

Theorem 2.5.1 (Surface area about the x -axis)

Suppose $f(x) \geq 0$ and f' is continuous on $[a, b]$. The area of the surface obtained by revolving $y = f(x)$ about the x -axis is

$$S = 2\pi \int_a^b f(x) \sqrt{1 + (f'(x))^2} dx.$$

If f may be negative, replace $f(x)$ by $|f(x)|$, since the radius is a distance and must be nonnegative.

The two factors in the integrand have clear geometric meanings:

$$\underbrace{2\pi f(x)}_{\text{circumference}} \underbrace{\sqrt{1 + (f'(x))^2} dx}_{\text{arc-length element}}.$$

Using only $2\pi f(x) dx$ would treat the sloping piece of the curve as horizontal and would therefore underestimate its surface area.

§2.5.ii. Rotation about the y -Axis

When $y = f(x)$ is revolved about the y -axis, the radius is the distance from the point $(x, f(x))$ to that axis. If $0 \leq a < b$, then $r = x$, and hence

$$S = 2\pi \int_a^b x \sqrt{1 + (f'(x))^2} dx.$$

More generally, if the interval includes negative values of x , the radius is $|x|$. Care must also be taken to avoid counting the same surface more than once if different parts of the curve trace the same surface after rotation.

If the curve is written as $x = g(y)$, where g' is continuous for $c \leq y \leq d$, then

$$ds = \sqrt{1 + (g'(y))^2} dy.$$

Consequently,

$$\text{about the } y\text{-axis:} \quad S = 2\pi \int_c^d |g(y)| \sqrt{1 + (g'(y))^2} dy,$$

$$\text{about the } x\text{-axis:} \quad S = 2\pi \int_c^d |y| \sqrt{1 + (g'(y))^2} dy.$$

§2.5.iii. Parametric Form

For a parametrized curve

$$x = x(t), \quad y = y(t), \quad \alpha \leq t \leq \beta,$$

the arc-length element is

$$ds = \sqrt{(x'(t))^2 + (y'(t))^2} dt.$$

Therefore, provided the surface is traced only once,

$$\begin{aligned} \text{about the } x\text{-axis:} \quad S &= 2\pi \int_{\alpha}^{\beta} |y(t)| \sqrt{(x'(t))^2 + (y'(t))^2} dt, \\ \text{about the } y\text{-axis:} \quad S &= 2\pi \int_{\alpha}^{\beta} |x(t)| \sqrt{(x'(t))^2 + (y'(t))^2} dt. \end{aligned}$$

In every form, the same principle is being used:

$$\boxed{dS = (\text{circumference}) ds = 2\pi(\text{radius}) ds.}$$

§2.6. Differential Equations

A *differential equation* is an equation involving an unknown function and one or more of its derivatives. Its *order* is the order of the highest derivative that occurs. For example,

$$y' = x - y \quad \text{and} \quad y'' + 4y = 0$$

are first- and second-order differential equations, respectively. A function is a solution on an interval only if it satisfies the equation at every point of that interval.

Most problems in elementary calculus are *initial-value problems*: the differential equation is accompanied by enough values of the unknown function and its derivatives to determine the arbitrary constants. A first-order equation normally requires one condition, such as $y(x_0) = y_0$, whereas a second-order equation normally requires two, such as $y(x_0) = y_0$ and $y'(x_0) = v_0$.

§2.6.i. First-Order Differential Equations

A general first-order equation has the form

$$F(x, y, y') = 0.$$

There is no single method that solves every such equation. The important first step is therefore to recognize its form.

Direction fields and Euler's method

If the equation can be written as

$$y' = f(x, y),$$

then the number $f(x, y)$ gives the slope of a solution curve through the point (x, y) . Drawing a short segment with this slope at many points produces a *direction field*. Even when an exact formula is unavailable, the field reveals whether solutions increase or decrease, where equilibrium solutions occur, and how solutions behave in the long run.

Euler's method replaces the solution curve by short tangent-line steps. Starting with (x_0, y_0) and using a step size h , define

$$x_{n+1} = x_n + h, \quad y_{n+1} = y_n + hf(x_n, y_n).$$

The approximation $y_n \approx y(x_n)$ is usually improved by choosing a smaller h , although more steps are then required.

The geometry of one Euler step is especially simple. At the current approximation $P_n = (x_n, y_n)$, the differential equation prescribes the slope $f(x_n, y_n)$. Moving horizontally by h along the corresponding tangent line changes the height by

$$\Delta y = (\text{slope})(\text{horizontal change}) = hf(x_n, y_n).$$

Thus the endpoint of that tangent-line segment is precisely $P_{n+1} = (x_n + h, y_n + hf(x_n, y_n))$.

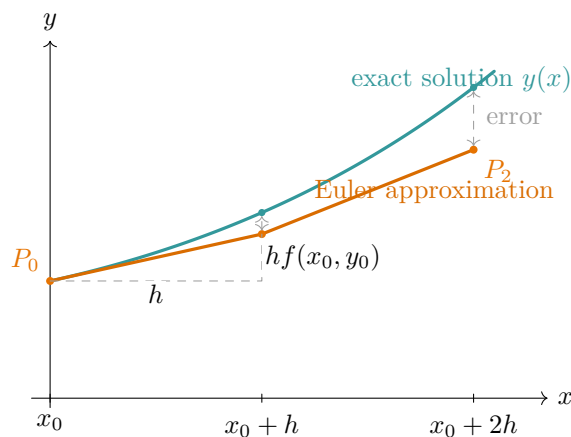


Figure 2.6.: Euler's method follows a tangent line for one short horizontal step and then recomputes the slope. The illustration uses $y' = x + y$, $y(0) = 1$, and $h = 0.4$; the larger step size makes the approximation error visible.

The orange segment from P_0 uses the slope at P_0 . At P_1 , Euler's method discards the old slope and uses the new slope $f(x_1, y_1)$ for the next segment. Notice that P_1 is generally not on the exact solution. Consequently, the next slope is also evaluated at an approximate point, so the error can accumulate from one step to the next. A smaller step size keeps each tangent-line segment closer to the curved solution.

Example 2.6.1 (Euler approximation)

For

$$y' = x + y, \quad y(0) = 1,$$

use $h = 0.1$. Since $f(x, y) = x + y$,

$$y_1 = 1 + 0.1f(0, 1) = 1.1.$$

The next step gives

$$y_2 = 1.1 + 0.1f(0.1, 1.1) = 1.22.$$

Thus Euler's method gives $y(0.2) \approx 1.22$.

Separable equations

An equation is *separable* if it can be written in the form

$$\frac{dy}{dx} = g(x)h(y).$$

When $h(y) \neq 0$, place all factors involving y with dy , place all factors involving x with dx , and integrate:

$$\frac{1}{h(y)} dy = g(x) dx, \quad \int \frac{1}{h(y)} dy = \int g(x) dx + C.$$

The result may define y implicitly. Before dividing by $h(y)$, one must separately check the constant solutions obtained from $h(y) = 0$.

Example 2.6.2 (A separable initial-value problem)

Solve

$$y' = xy, \quad y(0) = 3.$$

For a nonzero solution,

$$\frac{1}{y} dy = x dx.$$

Integration gives

$$\ln |y| = \frac{x^2}{2} + C',$$

so $y = Ce^{x^2/2}$, where the new constant C may be positive or negative. The initial condition yields $C = 3$. Therefore,

$$y = 3e^{x^2/2}.$$

The constant solution $y = 0$, lost when we divided by y , also solves the differential equation but not the given initial condition.

Two common models are immediate examples of separation. Exponential growth or decay satisfies

$$\frac{dP}{dt} = kP, \quad P(t) = P_0 e^{kt}.$$

Logistic growth satisfies

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{K} \right),$$

where $K > 0$ is the carrying capacity. If $P(0) = P_0 > 0$, then

$$P(t) = \frac{K}{1 + Ae^{-kt}}, \quad A = \frac{K - P_0}{P_0}.$$

First-order linear equations

A first-order equation is *linear* if it can be put into standard form

$$y' + P(x)y = Q(x).$$

Its integrating factor is

$$\mu(x) = e^{\int P(x) dx}.$$

Multiplying the entire equation by μ makes the left-hand side a product derivative:

$$\mu y' + \mu P y = (\mu y)' = \mu Q.$$

Consequently,

$$\mu(x)y(x) = \int \mu(x)Q(x) dx + C,$$

or

$$y(x) = \frac{1}{\mu(x)} \left(\int \mu(x)Q(x) dx + C \right).$$

Multiplying the equation by a nonzero constant multiple of μ gives the same family of solutions.

Example 2.6.3 (The integrating-factor method)

Solve

$$y' + 2y = e^x, \quad y(0) = 1.$$

Here $P(x) = 2$, so $\mu(x) = e^{2x}$. Multiplication gives

$$e^{2x}y' + 2e^{2x}y = e^{3x}, \quad (e^{2x}y)' = e^{3x}.$$

Hence

$$e^{2x}y = \frac{1}{3}e^{3x} + C, \quad y = \frac{1}{3}e^x + Ce^{-2x}.$$

From $y(0) = 1$, we obtain $C = 2/3$, and therefore

$$y = \frac{1}{3}e^x + \frac{2}{3}e^{-2x}.$$

Remark 2.6.1 (Choosing a first-order method). First simplify the equation and ask the following questions.

1. Can it be written as $y' = g(x)h(y)$? If so, separate the variables.
2. Can it be written as $y' + P(x)y = Q(x)$? If so, use an integrating factor.
3. If neither form applies, use a direction field or a numerical method to study the solution. Some equations require techniques beyond a first calculus course.

An equation may be both separable and linear; either valid method may be used.

§2.6.ii. Second-Order Differential Equations

A general second-order equation has the form

$$F(x, y, y', y'') = 0.$$

The main class treated in introductory calculus is the linear equation

$$a(x)y'' + b(x)y' + c(x)y = g(x).$$

It is *homogeneous* when $g(x) = 0$ and *nonhomogeneous* when $g(x) \neq 0$. The word homogeneous here refers to the right-hand side; it does not mean that all coefficients are constant.

Equations that can be reduced in order

Some second-order equations contain only one of the variables x and y . Their missing variable indicates a useful substitution that turns the equation into a first-order equation. After solving that first-order equation, one more integration recovers y . This process is called *reduction of order*.

Case 1: the dependent variable y is missing. Suppose the equation has the form

$$F(x, y', y'') = 0.$$

Because only derivatives of y occur, define a new unknown function

$$v(x) = y'(x).$$

Since v is a function of x ,

$$v'(x) = y''(x).$$

The original second-order equation therefore becomes the first-order equation

$$F(x, v, v') = 0.$$

The solution proceeds in two stages:

$\text{solve for } v \longrightarrow \text{solve } y' = v(x) \text{ for } y.$

The first stage introduces one constant, and the final integration normally introduces a second constant.

Example 2.6.4 (When y is missing)

Solve the initial-value problem

$$y'' + 3y' = 0, \quad y(0) = 2, \quad y'(0) = 6.$$

Set $v = y'$, so that $v' = y''$. The equation becomes

$$v' + 3v = 0.$$

This first-order equation is separable:

$$\frac{dv}{v} = -3 dx, \quad v = Ce^{-3x}.$$

Since $v(0) = y'(0) = 6$, we have $C = 6$. We have now found the derivative, not yet the original function:

$$y' = v = 6e^{-3x}.$$

Integrating once more gives

$$y = -2e^{-3x} + D.$$

Finally, $y(0) = 2$ implies $D = 4$. Hence

$y = 4 - 2e^{-3x}.$

Indeed, $y' = 6e^{-3x}$ and $y'' = -18e^{-3x}$, so $y'' + 3y' = 0$.

The especially simple equation $y'' = g(x)$ belongs to this case: integrate once to find y' , then integrate again to find y .

Case 2: the independent variable x is missing. Now suppose the equation has the form

$$F(y, y', y'') = 0.$$

Here the substitution is different. Regard $y' = v$ as a function of y , rather than directly as a function of x :

$$v(y) = y'.$$

The chain rule is essential. It gives

$$y'' = \frac{d}{dx}(v(y)) = \frac{dv}{dy} \frac{dy}{dx} = v \frac{dv}{dy}.$$

Thus make the replacements

$$y' = v, \quad y'' = v \frac{dv}{dy}.$$

The result is a first-order equation whose independent variable is now y . Solve it for $v(y)$, and then solve the first-order equation

$$\frac{dy}{dx} = v(y)$$

to recover y as a function of x .

Example 2.6.5 (When x is missing)

Solve

$$yy'' = (y')^2, \quad y(0) = 2, \quad y'(0) = 6.$$

Set $v(y) = y'$. Since $y'' = v \, dv/dy$, the equation becomes

$$yv \frac{dv}{dy} = v^2.$$

The initial value $v(2) = 6$ is nonzero, so near the initial point we may divide by yv :

$$\frac{1}{v} \frac{dv}{dy} = \frac{1}{y}.$$

Integrating with respect to y gives

$$\ln |v| = \ln |y| + C, \quad v = Ky.$$

Since $v(2) = 6$, we obtain $K = 3$. Returning to the original variables now gives a first-order equation:

$$y' = v = 3y.$$

Its solution is $y = Ae^{3x}$, and $y(0) = 2$ gives $A = 2$. Therefore,

$$y = 2e^{3x}.$$

Remark 2.6.2 (Two substitutions that must not be confused). When y is missing, use $v(x) = y'$ and hence $y'' = v'$. When x is missing, use $v(y) = y'$ and hence $y'' = v \, dv/dy$. Writing $y'' = v'$ in the second case without specifying the variable of differentiation loses the chain-rule factor v .

As with separable equations, check separately for solutions lost by division. For example, dividing by v can discard the branch $v = 0$, which corresponds to constant solutions $y' = 0$.

Homogeneous equations with constant coefficients

Consider

$$ay'' + by' + cy = 0, \quad a \neq 0,$$

where a, b, c are real constants. It is called *homogeneous* because its right-hand side is zero. Before trying to solve it, we first explain the structure of its set of solutions.

Theorem 2.6.6 (Existence and uniqueness for the initial-value problem)

Fix $x_0 \in \mathbb{R}$. For every pair of real numbers A, B , the problem

$$ay'' + by' + cy = 0, \quad y(x_0) = A, \quad y'(x_0) = B$$

has exactly one solution.

Proof. Divide the equation by a and write it as

$$y'' = py' + qy, \quad p = -\frac{b}{a}, \quad q = -\frac{c}{a}.$$

We prove existence and uniqueness separately, without using the characteristic-root formulas that will be derived later.

Existence. Seek a power-series solution centered at x_0 :

$$y(x) = \sum_{n=0}^{\infty} c_n(x - x_0)^n.$$

The initial conditions require

$$c_0 = A, \quad c_1 = B.$$

Formally differentiating gives

$$\begin{aligned} y' &= \sum_{n=0}^{\infty} (n+1)c_{n+1}(x - x_0)^n, \\ y'' &= \sum_{n=0}^{\infty} (n+2)(n+1)c_{n+2}(x - x_0)^n. \end{aligned}$$

Matching the coefficient of $(x - x_0)^n$ in $y'' = py' + qy$ gives the recurrence

$$(n+2)(n+1)c_{n+2} = p(n+1)c_{n+1} + qc_n.$$

Thus $c_0 = A$ and $c_1 = B$ determine c_2 , which determines c_3 , and so on. There is therefore at most one power series with the prescribed initial values.

We must verify that the resulting series actually converges. Choose $R \geq 1$ large enough that

$$R^2 \geq |p|R + |q|,$$

and let

$$K = \max \left\{ |A|, \frac{|B|}{R} \right\}.$$

We claim that

$$|c_n| \leq K \frac{R^n}{n!} \quad (n \geq 0).$$

This is true for $n = 0, 1$. If it holds for n and $n + 1$, the recurrence gives

$$\begin{aligned} |c_{n+2}| &\leq \frac{|p|(n+1)|c_{n+1}| + |q||c_n|}{(n+2)(n+1)} \\ &\leq K \frac{R^n(|p|R + |q|)}{(n+2)!} \\ &\leq K \frac{R^{n+2}}{(n+2)!}. \end{aligned}$$

Induction proves the claim.

Consequently, for every real x ,

$$\sum_{n=0}^{\infty} |c_n| |x - x_0|^n \leq K \sum_{n=0}^{\infty} \frac{(R|x - x_0|)^n}{n!} = K e^{R|x - x_0|}.$$

The series therefore converges for every x . The same estimate applied after one or two differentiations shows that the derivative series also converge on every bounded interval. Hence the series may be differentiated twice term by term. The recurrence then guarantees that the resulting function satisfies $y'' = py' + qy$, while $c_0 = A$ and $c_1 = B$ give the prescribed initial values. A solution therefore exists. ■

Uniqueness. Suppose y_1 and y_2 are two solutions with the same initial values, and set

$$w = y_1 - y_2.$$

Then

$$w'' = pw' + qw, \quad w(x_0) = w'(x_0) = 0.$$

Define the nonnegative function

$$E(x) = w(x)^2 + w'(x)^2.$$

Using the differential equation,

$$\begin{aligned} E' &= 2ww' + 2w'w'' \\ &= 2(1+q)ww' + 2p(w')^2. \end{aligned}$$

Since $2|ww'| \leq w^2 + (w')^2 = E$, there is a constant

$$M = |1+q| + 2|p|$$

such that

$$|E'(x)| \leq ME(x).$$

For $x \geq x_0$, the inequality $E' \leq ME$ gives

$$\frac{d}{dx}(e^{-Mx}E(x)) \leq 0.$$

Since $E(x_0) = 0$ and $E \geq 0$, this forces $E(x) = 0$ for $x \geq x_0$. For $x \leq x_0$, the inequality $E' \geq -ME$ similarly gives

$$\frac{d}{dx}(e^{Mx}E(x)) \geq 0,$$

which again forces $E(x) = 0$. Thus $E \equiv 0$, so $w \equiv 0$, and consequently $y_1 = y_2$. The solution is unique. ■

The proof shows precisely why two initial values may be chosen freely: they are the first two coefficients of the power series, and the differential equation recursively determines every later coefficient.

Theorem 2.6.7 (The solution space has dimension two)

Let I be an interval containing x_0 , and let

$$\mathcal{S} = \{y \in C^2(I, \mathbb{R}) : ay'' + by' + cy = 0\}.$$

Then $\mathcal{S}$ is a two-dimensional real vector space.

Proof. The zero function is a solution. Moreover, if y_1, y_2 are solutions and $s, t \in \mathbb{R}$, then

$$\begin{aligned} a(sy_1 + ty_2)'' + b(sy_1 + ty_2)' + c(sy_1 + ty_2) &= s(ay_1'' + by_1' + cy_1) \\ &\quad + t(ay_2'' + by_2' + cy_2) \\ &= 0. \end{aligned}$$

Thus every linear combination of solutions is again a solution, so $\mathcal{S}$ is a vector space.

Now assign to each solution its two initial values:

$$\Phi : \mathcal{S} \longrightarrow \mathbb{R}^2, \quad \Phi(y) = (y(x_0), y'(x_0)).$$

This map is linear. The existence part of the preceding theorem says that every pair $(A, B) \in \mathbb{R}^2$ is $\Phi(y)$ for some solution y , so Φ is onto. The uniqueness part says that two solutions with the same initial values must be identical, so Φ is one-to-one. Hence Φ is a linear isomorphism

$$\mathcal{S} \cong \mathbb{R}^2.$$

Since $\mathbb{R}^2$ has dimension 2, so does $\mathcal{S}$. ■

This theorem gives the central strategy. We do not need to guess every solution directly. It is enough to find two solutions y_1, y_2 that are *linearly independent*, meaning that

$$C_1y_1 + C_2y_2 = 0 \quad \text{as a function}$$

is possible only when $C_1 = C_2 = 0$. Two linearly independent vectors in a two-dimensional vector space form a basis. Therefore every solution is then automatically of the form

$$\boxed{y = C_1y_1 + C_2y_2.}$$

Finding special solutions with an exponential. Exponential functions are useful because differentiation does not change their basic form:

$$y = e^{rx}, \quad y' = re^{rx}, \quad y'' = r^2e^{rx}.$$

Substitution gives

$$ay'' + by' + cy = (ar^2 + br + c)e^{rx}.$$

Because $e^{rx} \neq 0$, the function e^{rx} is a solution precisely when r satisfies the *characteristic equation*

$$ar^2 + br + c = 0.$$

We now use its roots to find two independent special solutions.

Case 1: two distinct real roots. Suppose the characteristic roots are $r_1 \neq r_2$. Then

$$y_1 = e^{r_1x}, \quad y_2 = e^{r_2x}$$

are solutions. To check independence, suppose

$$C_1e^{r_1x} + C_2e^{r_2x} = 0$$

for every x . Evaluating this equation and its derivative at x_0 gives

$$\begin{aligned} C_1e^{r_1x_0} + C_2e^{r_2x_0} &= 0, \\ r_1C_1e^{r_1x_0} + r_2C_2e^{r_2x_0} &= 0. \end{aligned}$$

Subtracting r_1 times the first equation from the second gives

$$(r_2 - r_1)C_2e^{r_2x_0} = 0.$$

Thus $C_2 = 0$, and then $C_1 = 0$. The two special solutions are independent. Since $\dim \mathcal{S} = 2$, they form a basis, and hence

$$\boxed{y = C_1e^{r_1x} + C_2e^{r_2x}.}$$

Case 2: one repeated real root. Suppose the characteristic equation has the repeated root r_0 . The exponential method initially gives only one solution,

$$y_1 = e^{r_0 x}.$$

We need a second one. Since the characteristic polynomial is $a(r - r_0)^2$, its coefficients satisfy

$$b = -2ar_0, \quad c = ar_0^2.$$

Set $y = e^{r_0 x}u(x)$. Then

$$y' = e^{r_0 x}(r_0 u + u'), \quad y'' = e^{r_0 x}(r_0^2 u + 2r_0 u' + u'').$$

Substitution into $ay'' + by' + cy = 0$ cancels every term except

$$ae^{r_0 x}u'' = 0.$$

Thus $u'' = 0$. Choosing $u = 1$ and $u = x$ gives the two special solutions

$$y_1 = e^{r_0 x}, \quad y_2 = xe^{r_0 x}.$$

They are independent: if

$$C_1 e^{r_0 x} + C_2 x e^{r_0 x} = 0,$$

then division by the nonzero function $e^{r_0 x}$ gives $C_1 + C_2 x = 0$ for every x , which forces $C_1 = C_2 = 0$. Therefore they form a basis of the two-dimensional solution space, and

$$\boxed{y = (C_1 + C_2 x)e^{r_0 x}}.$$

Case 3: a complex-conjugate pair. Suppose the characteristic roots are

$$r = \alpha \pm i\beta, \quad \beta > 0.$$

Euler's identity

$$e^{i\beta x} = \cos(\beta x) + i \sin(\beta x)$$

leads to the two real special solutions

$$y_1 = e^{\alpha x} \cos(\beta x), \quad y_2 = e^{\alpha x} \sin(\beta x).$$

They can also be verified directly by substitution. To check independence, suppose $C_1 y_1 + C_2 y_2 = 0$. Dividing by $e^{\alpha x}$ gives

$$C_1 \cos(\beta x) + C_2 \sin(\beta x) = 0.$$

Differentiating gives

$$-\beta C_1 \sin(\beta x) + \beta C_2 \cos(\beta x) = 0.$$

At any fixed x , these two equations imply $C_1 = C_2 = 0$, because $\cos^2(\beta x) + \sin^2(\beta x) = 1$ and $\beta \neq 0$. The two solutions are therefore independent and form a basis. Hence

$$\boxed{y = e^{\alpha x} (C_1 \cos(\beta x) + C_2 \sin(\beta x))}.$$

Remark 2.6.3 (The logical structure of the method). The characteristic equation finds two special solutions. The existence-and-uniqueness theorem tells us beforehand that the solution space has dimension 2. After checking that the two special solutions are linearly independent, linear algebra tells us that they form a basis; only at that point may we conclude that their linear combination is the general solution.

Example 2.6.8 (A second-order initial-value problem)

Solve

$$y'' - 3y' + 2y = 0, \quad y(0) = 4, \quad y'(0) = 1.$$

The characteristic equation is

$$r^2 - 3r + 2 = (r - 1)(r - 2) = 0,$$

so

$$y = C_1 e^x + C_2 e^{2x}, \quad y' = C_1 e^x + 2C_2 e^{2x}.$$

The initial conditions give

$$C_1 + C_2 = 4, \quad C_1 + 2C_2 = 1.$$

Thus $C_2 = -3$ and $C_1 = 7$, so

$$y = 7e^x - 3e^{2x}.$$

Nonhomogeneous equations with constant coefficients

For

$$ay'' + by' + cy = g(x),$$

the general solution has the form

$$y = y_h + y_p,$$

where y_h is the general solution of the associated homogeneous equation and y_p is any one particular solution of the nonhomogeneous equation.

When $g(x)$ is a polynomial, an exponential, a sine or cosine, or a sum or product of these, a particular solution can often be found by *undetermined coefficients*. Choose a trial function with the same basic form as g , substitute it into the equation, and solve for its unknown coefficients. If any term in the trial function already occurs in y_h , multiply the entire trial by enough powers of x to make it linearly independent of the homogeneous solutions.

Forcing term $g(x)$	Typical trial for y_p
polynomial of degree n	general polynomial of degree n
e^{kx}	Ae^{kx}
$\cos(kx)$ or $\sin(kx)$	$A \cos(kx) + B \sin(kx)$
sum of terms	sum of the corresponding trials

Example 2.6.9 (Undetermined coefficients)

Solve

$$y'' - y = 2e^{2x}.$$

The homogeneous characteristic equation $r^2 - 1 = 0$ gives

$$y_h = C_1 e^x + C_2 e^{-x}.$$

Since e^{2x} is not a homogeneous solution, try $y_p = Ae^{2x}$. Then

$$y_p'' - y_p = (4A - A)e^{2x} = 3Ae^{2x}.$$

Matching this with $2e^{2x}$ gives $A = 2/3$. Therefore,

$$y = C_1 e^x + C_2 e^{-x} + \frac{2}{3} e^{2x}.$$

Example 2.6.10 (Resonance)

Solve

$$y'' + y = \cos x.$$

The homogeneous solution is

$$y_h = C_1 \cos x + C_2 \sin x.$$

The usual trial $A \cos x + B \sin x$ duplicates y_h , so multiply it by x . A convenient equivalent trial is $y_p = Ax \sin x$. Substitution gives

$$y_p'' + y_p = 2A \cos x,$$

and hence $A = 1/2$. Thus

$$y = C_1 \cos x + C_2 \sin x + \frac{x}{2} \sin x.$$

Oscillation models

The ideal spring equation

$$my'' + ky = 0, \quad m, k > 0,$$

has angular frequency $\omega = \sqrt{k/m}$ and solution

$$y = C_1 \cos(\omega t) + C_2 \sin(\omega t) = A \cos(\omega t - \delta).$$

Adding damping and an external force gives

$$my'' + cy' + ky = F(t).$$

The characteristic roots classify the unforced motion as overdamped, critically damped, or underdamped. A periodic force can create resonance when its frequency is near the natural frequency.

To solve a differential equation, first classify it. Separate variables when the x - and y -dependence can be separated; use an integrating factor for a first-order linear equation; and use the characteristic equation for a second-order linear equation with constant coefficients. Find the general solution before applying the initial conditions, and always substitute the final answer back into the differential equation to check it.